

$$x^2 + (3k - 7)x + (k^2 - 3k) = 0$$

$$\Delta = (3k - 7)^2 - 4(k^2 - 3k)(1)$$

$$= 9k^2 - 42k + 49 - 4k^2 + 12k$$

$$= 5k^2 - 30k + 49$$

$$= 5(k - 3)^2 + 49 - 45$$

$$= 5(k - 3)^2 + 4$$

The discriminant is greater than zero for all real k because it is a parabola that is always above the x -axis (positive value), so the equation has two distinct real roots for all real values of k .